ON A PROPER CURVE WITH H^0(X, O_X) = K, A GLOBALLY GENERATED-IN-PRINCIPLE INVERTIBLE SHEAF WITH A REGULAR SECTION, VANISHING H^1 AND AMPLENESS HAS VERY AMPLE SIXTH TENSOR POWER

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the Cartier zero scheme setup with the author's small-detail remark on nonemptiness, the induction giving vanishing and trivializations, the two-term multiplication map surjectivity, and the closing Veronese generation argument are all retained verbatim.

Editorial note. The label lemma-criterion-very-ample names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Zero schemes of regular sections. Divisors, Section divisors-section-effective-Cartier-invertible, Lemma divisors-lemma-effective-Cartier-makes-dimension-drop; Varieties, Lemma varieties-lemma-algebraic-scheme-dim-0. Supplies the Cartier zero scheme Z of the regular section, its dimension 0, and the exact sequences 0 -> L^tensor n -> L^tensor n+1 -> i_*(L^tensor n+1|_Z) -> 0.
- Proj of a section ring generated in bounded degree. Morphisms, Lemma morphisms-lemma-proper-ample-is-proj; Constructions, Lemma constructions-lemma-surjective-graded-rings-generated-degree-1-map-proj. Identifies X = Proj(S) for S the section ring and converts generation of the Veronese S^(6) in degree 1 into very-ampleness of L^tensor 6.

Source reference. Original lemma 02M9: sources/stacks-chow/tags.txt, source lines 3436–3436. Prerequisite pointer invoked at proof line(s) 1538; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 056N: sources/stacks-chow/tags.txt, source lines 6716–6716. Prerequisite pointer invoked at proof line(s) 1534; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0C4S: sources/stacks-chow/tags.txt, source lines 15237–15237. Prerequisite pointer invoked at proof line(s) 1521; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0C6J: sources/stacks-chow/tags.txt, source lines 15299–15299. Prerequisite pointer invoked at proof line(s) 1588; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06LH: sources/stacks-chow/tags.txt, source lines 8466–8466. Prerequisite pointer invoked at proof line(s) 1536; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Prerequisite pointer invoked at proof line(s) 1541; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume*

(1) *$\mathcal{L}$ has a regular global section,*
(2) *$H^1(X, \mathcal{L}) = 0$, and*
(3) *$\mathcal{L}$ is ample.*

Then $\mathcal{L}^{\otimes 6}$ is very ample on X over k.

Proof. Let s be a regular global section of $\mathcal{L}$. Let $i : Z = Z(s) \to X$ be the zero scheme of s, see Divisors, Section 0C4S. By condition (3) we see that $Z \neq \emptyset$ (small detail omitted). Consider the short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} \mathcal{L} \to i_*(\mathcal{L}|_Z) \to 0$$

Tensoring with $\mathcal{L}$ we obtain

$$0 \to \mathcal{L} \to \mathcal{L}^{\otimes 2} \to i_*(\mathcal{L}^{\otimes 2}|_Z) \to 0$$

Observe that Z has dimension 0 (Divisors, Lemma 056N) and hence is the spectrum of an Artinian ring (Varieties, Lemma 06LH) hence $\mathcal{L}|_Z \cong \mathcal{O}_Z$ (Algebra, Lemma 02M9). The short exact sequence also shows that $H^1(X, \mathcal{L}^{\otimes 2}) = 0$ (for example using Varieties, Lemma 0AYT to see vanishing in the spot on the right). Using induction on $n \geq 1$ and the sequence

$$0 \to \mathcal{L}^{\otimes n} \xrightarrow{s} \mathcal{L}^{\otimes n+1} \to i_*(\mathcal{L}^{\otimes n+1}|_Z) \to 0$$

we see that $H^1(X, \mathcal{L}^{\otimes n}) = 0$ for $n > 0$ and that there exists a global section t_{n+1} of $\mathcal{L}^{\otimes n+1}$ which gives a trivialization of $\mathcal{L}^{\otimes n+1}|_Z \cong \mathcal{O}_Z$.

Consider the multiplication map

$$\mu_n : H^0(X, \mathcal{L}) \otimes_k H^0(X, \mathcal{L}^{\otimes n}) \oplus H^0(X, \mathcal{L}^{\otimes 2}) \otimes_k H^0(X, \mathcal{L}^{\otimes n-1}) \longrightarrow H^0(X, \mathcal{L}^{\otimes n+1})$$

We claim this is surjective for $n \geq 3$. To see this we consider the short exact sequence

$$0 \to \mathcal{L}^{\otimes n} \xrightarrow{s} \mathcal{L}^{\otimes n+1} \to i_*(\mathcal{L}^{\otimes n+1}|_Z) \to 0$$

The sections of $\mathcal{L}^{\otimes n+1}$ coming from the left in this sequence are in the image of μ_n. On the other hand, since $H^0(\mathcal{L}^{\otimes 2}) \to H^0(\mathcal{L}^{\otimes 2}|_Z)$ is surjective (see above) and since t_{n-1} maps to a trivialization of $\mathcal{L}^{\otimes n-1}|_Z$ we see that $\mu_n(H^0(X, \mathcal{L}^{\otimes 2}) \otimes t_{n-1})$ gives a subspace of $H^0(X, \mathcal{L}^{\otimes n+1})$ surjecting onto the global sections of $\mathcal{L}^{\otimes n+1}|_Z$. This proves the claim.

From the claim in the previous paragraph we conclude that the graded k-algebra

$$S = \bigoplus\nolimits_{n \geq 0} H^0(X, \mathcal{L}^{\otimes n})$$

is generated in degrees $0, 1, 2, 3$ over k. Recall that $X = \mathrm{Proj}(S)$, see Morphisms, Lemma 0C6J. Thus $S^{(6)} = \bigoplus_n S_{6n}$ is generated in degree 1. This means that $\mathcal{L}^{\otimes 6}$ is very ample as desired. $\square$